中国船级社
CHINA CLASSIFICATION SOCIETY
船用产品证书
CERTIFICATE OF MARINE PRODUCT

证书编号/Certificate No.
ZG26PPS01017

兹证明本证书所列产品经本社署名验船师检验，符合本证书注明标准的要求。
This is to certify that the following products have been inspected by the undersigned surveyor of the Society and are found to comply with the requirements of the specified standards.

产品名称 Product	**铜合金螺旋桨 Copper Alloy Propeller**
申请方 Applicant	浙江元鼎船舶设备有限公司 Zhoushan East Ocean Propeller-Making Co.,Ltd.
制造厂 Manufacturer	浙江元鼎船舶设备有限公司 Zhoushan East Ocean Propeller-Making Co.,Ltd.
订货方 Purchaser	杭州前进齿轮箱集团股份有限公司 Barge Hengue 019

认可证书号/Certificate No. of Approval ZG26PWA00002 图纸批准号/Approval No. of Drawings 详见附页/See additional page(s)

用于/Intended for 国际航行海船/International sea-going ship
船舶工程号/Hull No.:VHX9158

产品编号/Serial No. **详见附页/See additional page(s)**

产品检验标准/Product Inspection Standard

1.中国船级社《材料与焊接规范》（2025）及其变更通告第1篇第9章
Chapter 9, Part One of China Classification Society Rules for Materials and Welding 2025 and its Change Notices

产品明细/Product Description

铜合金螺旋桨/Copper Alloy Propeller(M0001)

名称/Name	属性（值）/Value	单位/Unit
结构/Structure	详见附页/See additional page(s)	
合金类型/Alloy Types	详见附页/See additional page(s)	
数量/Quantity	详见附页/See additional page(s)	
炉号/Cast No.	详见附页/See additional page(s)	
无损检测/Non-Destructive Test	详见附页/See additional page(s)	
图号/Drawing No.	详见附页/See additional page(s)	
直径/Diameter	详见附页/See additional page(s)	
叶片数量/Number of Blade	详见附页/See additional page(s)	
旋转方向/Direction of Rotation	详见附页/See additional page(s)	
重量/Weight	详见附页/See additional page(s)	

产品检验标志/Marking

位置/Position : 在产品上 / On the product
标志样式 Type of Marking : ZG26PPS01017 钢印标志 / CCS steel stamp

发证日期 Date of issue 2026年02月27日 Feb. 27, 2026

验船师 Surveyor Wang Pingzong

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This Certificate is issued pursuant to the Rules of the Society and related regulation. All pages of the certificate are taken as a whole and are used simultaneously. No paper certificate page is valid without bearing the stamp of the Society, no electronic certificates is valid without the digital signature, and no copied form of the certificate is regarded as valid. Any part of the certificate is not to be extracted or abridged by any unit or individual in any form. Related parties who are doubted about the authenticity of the certificate may inquire of the Society or its offices.

备注/Remarks

无/Nil.

中国船级社浙江分社
CCS Zhejiang Branch
注：本证书含有附页，共2页
Note: The certificate is attached with additional 2 page(s)

产品明细/Product Description
铜合金螺旋桨/Copper Alloy Propeller（M0001）

名称/Name	属性（值）/Value	单位/Unit
结构/Structure	整体式/Integral type	
数量/Quantity	1	件/Pcs
产品编号/Serial No.	DH26010021-1	
叶片数量/Number of Blade	4	片/Pcs
重量/Weight	697	Kg
图号/Drawing No.	HS9266-425-017	
图纸批准号/Approval No. of Drawings	WP26SNP00026-003H	
表面状态/Surface Machining	精加工/Finished machining	
无损检测/Non-Destructive Test	着色探伤/PT	
炉号/Cast No.	2405163	
合金类型/Alloy Types	CU3	
0.7R 处螺距/Pitch at 0.7r	1230	mm
侧斜角/Side Bevel	5.44	度/degree
静平衡实验/Static Balance Test	合格/Qualified	
直径/Diameter	2020	mm
旋转方向/Direction of Rotation	左旋/Left ration	
断后伸长率/Percentage Elongation After Fracture（A5）	19%	
抗拉强度/Tensile Strength（Rm）	620	MPa
规定非比例延伸强度/Proof Strength（Rp0.2）	305	MPa
化学成分/Chemical Composition	Cu：80.01%, Al：8.92% , Mn：1.03% , Zn：0.35% , Fe：4.97% , Ni：4.09%, Sn：0.07%, Pb：0.03% , Si：0.056%	

中国船级社浙江分社 CCS Zhejiang Branch

铜合金螺旋桨/Copper Alloy Propeller（M0002）

名称/Name	属性（值）/Value	单位/Unit
结构/Structure	整体式/Integral type	
数量/Quantity	1	件/Pcs
产品编号/Serial No.	DH26010021-2	
叶片数量/Number of Blade	4	片/Pcs
重量/Weight	701	Kg
图号/Drawing No.	HS9266-425-017	
图纸批准号/Approval No. of Drawings	WP23SNP00026-003H	
表面状态/Surface Machining	精加工/Finished machining	
无损检测/Non-Destructive Test	着色探伤/PT	
炉号/Cast No.	2405152	
合金类型/Alloy Types	CU3	
0.7R 处螺距/Pitch at 0.7r	1230	mm
侧斜角/Side Bevel	5.44	度/degree
静平衡实验/Static Balance Test	合格/Qualified	
直径/Diameter	2020	mm
旋转方向/Direction of Rotation	右旋/Right ration	
断后伸长率/Percentage Elongation After Fracture（A5）	19%	
抗拉强度/Tensile Strength（Rm）	620	MPa
规定非比例延伸强度/Proof Strength（Rp0.2）	305	MPa
化学成分/Chemical Composition	Cu：80.01%, Al：8.92% , Mn：1.03% , Zn：0.35% , Fe：4.97% , Ni：4.09%, Sn：0.07%, Pb：0.03% , Si：0.057%	

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